

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION**CLASS IX • SOCIAL SCIENCE (SUBJECT CODE: 087) • ACADEMIC SESSION 2026-27****SET A — FOUNDATION & CORE CONCEPTS****CHAPTER 02: SHAPING OF THE EARTH'S SURFACE**

TIME ALLOWED: 45 MINUTES

MAXIMUM MARKS: 25 MARKS

SERIES: CBS-IX-SST-SET-A-02

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

General Instructions (SET A - Foundation & Core Concepts):

1. The question paper comprises **25 Multiple Choice Questions** of **1 mark each**.
2. All questions are compulsory. There is no negative marking for incorrect answers.
3. Each question offers four options: **(A), (B), (C), and (D)**. Select the single best answer.
4. Darken the corresponding circle on the **OMR Sheet** completely using a blue/black ballpoint pen.

Q.1 Forces that originate within the Earth's interior and cause movements of the crust (such as earthquakes and volcanic eruptions) are known as:

- | | |
|------------------------|-------------------------|
| (A) Exogenic forces | (B) Endogenic forces |
| (C) Atmospheric forces | (D) Gravitational tides |

Q.2 The theory that Earth's lithosphere is broken into several large and small rigid slabs that float upon the semi-fluid asthenosphere is called:

- | | |
|---------------------------|----------------------------|
| (A) Plate Tectonics | (B) Continental Inundation |
| (C) Isostatic Equilibrium | (D) Solar Radiation Theory |

Q.3 When two tectonic plates move toward each other and collide, leading to subduction or mountain building, the boundary is classified as:

- | | |
|------------------------------|-------------------------|
| (A) Divergent boundary | (B) Convergent boundary |
| (C) Transform fault boundary | (D) Stationary boundary |

Q.4 The Mid-Atlantic Ridge is a classic global example of which type of plate tectonic boundary?

- | | |
|------------------------------|------------------------|
| (A) Convergent boundary | (B) Divergent boundary |
| (C) Transform fault boundary | (D) Subduction zone |

Q.5 The point deep inside the Earth's crust where seismic energy is initially released during an earthquake is termed the:

- | | |
|-----------------|------------------------|
| (A) Epicentre | (B) Focus (Hypocentre) |
| (C) Fault scarp | (D) Crater |

Q.6 The point on the surface of the Earth located vertically directly above the earthquake's underground focus is known as the:

- | | |
|------------------|-------------|
| (A) Epicentre | (B) Crater |
| (C) Seismic core | (D) Caldera |

Q.7 Earthquake tremors and seismic waves are scientifically detected and recorded using an instrument called a:

- | | |
|----------------|-----------------|
| (A) Barometer | (B) Seismograph |
| (C) Anemometer | (D) Hygrometer |

Q.8 The magnitude of energy released by an earthquake is measured on which logarithmic scale?

- | | |
|--------------------|-------------------|
| (A) Beaufort scale | (B) Richter scale |
| (C) Kelvin scale | (D) Mohs scale |

Q.9 The circular opening or vent at the summit of a volcano through which magma, gases, and ash erupt is called the:

- | | |
|------------|------------|
| (A) Crater | (B) Gorge |
| (C) Cirque | (D) Canyon |

Q.10 The disintegration and decomposition of rocks in situ (in place) without significant transport by agents of movement is known as:

- | | |
|----------------|--------------------|
| (A) Weathering | (B) Erosion |
| (C) Deposition | (D) Transportation |

Q.11 Which of the following processes involves both the detachment and active transport of weathered rock materials by moving agents like water, wind, or ice?

- | | |
|-------------------|-------------------|
| (A) Weathering | (B) Erosion |
| (C) Sedimentation | (D) Lithification |

Q.12 A deep, narrow valley with extremely steep, nearly vertical rocky walls carved by a vigorous, swift-flowing young river is known as a:

- | | |
|--------------------|------------------|
| (A) Gorge / Canyon | (B) Oxbow lake |
| (C) Delta | (D) Alluvial fan |

Q.13 As a river leaves the mountains and flows across a broad, flat plain, it forms large curving loops and broad bends known as:

- | | |
|----------------|--------------|
| (A) Rapids | (B) Meanders |
| (C) Waterfalls | (D) Moraines |

Q.14 When continuous erosion and deposition along the neck of a river meander eventually causes the river to cut straight through, the abandoned crescent loop forms an:

- | | |
|----------------|------------------|
| (A) Oxbow lake | (B) Alluvial fan |
| (C) Estuary | (D) Oasis |

Q.15 The triangular or fan-shaped tract of fertile sediment deposited at the mouth of a river where it enters a sea or ocean is called a:

- | | |
|-----------|---------------|
| (A) Delta | (B) Cirque |
| (C) Gorge | (D) Sand dune |

Q.16 Which of the following is a characteristic landform carved by alpine or mountain glacial erosion?

- | | |
|---------------------|---------------------|
| (A) U-shaped valley | (B) V-shaped valley |
| (C) Mushroom rock | (D) Sand dune |

Q.17 The ridges of unsorted rock debris, boulders, gravel, and sand transported and left behind by melting glaciers are called:

- | | |
|--------------|------------|
| (A) Moraines | (B) Loess |
| (C) Deltas | (D) Levees |

Q.18 Isolated rock pedestals in arid deserts shaped by abrasive wind erosion that wears down softer rock at the base faster than the top are called:

- | | |
|--------------------|----------------|
| (A) Mushroom rocks | (B) Barchans |
| (C) Oxbows | (D) Sea stacks |

Q.19 Crescent-shaped sand dunes formed in desert regions with windward slopes facing the wind and tips pointing downwind are called:

- | | |
|--------------------|------------------|
| (A) Barchans | (B) Loess plains |
| (C) Drift terraces | (D) Moraines |

Q.20 Extensive, fertile deposits of fine, windblown yellowish dust carried across hundreds of kilometers (such as in northern China) are called:

- | | |
|-----------|--------------|
| (A) Loess | (B) Gravel |
| (C) Till | (D) Laterite |

Q.21 Steep, nearly vertical rocky cliffs facing the sea formed by continuous wave erosion at the base are termed:

- | | |
|----------------|-------------|
| (A) Sea cliffs | (B) Beaches |
| (C) Sand dunes | (D) Deltas |

Q.22 When wave erosion cuts completely through a coastal headland from both sides, it forms a tunnel-like arch known as a:

- | | |
|--------------|---------------|
| (A) Sea arch | (B) Sea stack |
| (C) Sea cave | (D) Spit |

Q.23 When the roof of a sea arch eventually collapses under gravity, the remaining isolated vertical rock pillar standing in the sea is called a:

- | | |
|---------------|-------------------|
| (A) Sea stack | (B) Mushroom rock |
| (C) Spit | (D) Gorge |

Q.24 The low, gently sloping landform along a coastline composed of loose sand, gravel, and shell fragments deposited by constructive waves is a:

- | | |
|-------------|------------|
| (A) Beach | (B) Gorge |
| (C) Moraine | (D) Crater |

Q.25 Which plate boundary interaction is responsible for the formation of the Himalayan mountain range?

- | | |
|---|---|
| (A) Collision between the Indian Plate and the Eurasian Plate (Convergent Boundary) | (B) Pulling apart of the African and South American Plates |
| (C) Sliding past of the Pacific and North American Plates along San Andreas Fault | (D) Sinking of the Australian continent into the Indian Ocean |

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET A)

01	☐ A ☐ B ☐ C ☐ D	02	☐ A ☐ B ☐ C ☐ D	03	☐ A ☐ B ☐ C ☐ D	04	☐ A ☐ B ☐ C ☐ D	05	☐ A ☐ B ☐ C ☐ D
06	☐ A ☐ B ☐ C ☐ D	07	☐ A ☐ B ☐ C ☐ D	08	☐ A ☐ B ☐ C ☐ D	09	☐ A ☐ B ☐ C ☐ D	10	☐ A ☐ B ☐ C ☐ D
11	☐ A ☐ B ☐ C ☐ D	12	☐ A ☐ B ☐ C ☐ D	13	☐ A ☐ B ☐ C ☐ D	14	☐ A ☐ B ☐ C ☐ D	15	☐ A ☐ B ☐ C ☐ D
16	☐ A ☐ B ☐ C ☐ D	17	☐ A ☐ B ☐ C ☐ D	18	☐ A ☐ B ☐ C ☐ D	19	☐ A ☐ B ☐ C ☐ D	20	☐ A ☐ B ☐ C ☐ D
21	☐ A ☐ B ☐ C ☐ D	22	☐ A ☐ B ☐ C ☐ D	23	☐ A ☐ B ☐ C ☐ D	24	☐ A ☐ B ☐ C ☐ D	25	☐ A ☐ B ☐ C ☐ D

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & STEP-BY-STEP EXPLANATIONS (SET A)

Q. No.	Correct	Curricular Reference & Conceptual Rationale
Q.1	(B)	Endogenic forces are internal forces originating deep within the Earth, responsible for mountain building (orogeny), volcanism, and earthquakes.
Q.2	(A)	Plate Tectonics explains the structural features of the Earth's crust in terms of the interaction of rigid lithospheric plates moving over the asthenosphere.
Q.3	(B)	At a convergent boundary, two plates collide. One plate is forced beneath the other (subduction) or crust crumples upward into fold mountains (e.g., Himalayas).
Q.4	(B)	At a divergent boundary, tectonic plates pull apart and molten magma rises to form new oceanic crust, as seen along the Mid-Atlantic Ridge.
Q.5	(B)	The focus (hypocentre) is the actual underground point of origin of an earthquake. The point on the Earth's surface directly above the focus is the epicentre.
Q.6	(A)	The epicentre is the location on the Earth's surface directly above the focus where earthquake vibrations are typically felt first and most intensely.
Q.7	(B)	A seismograph (or seismometer) records the vibrations and seismic waves generated by earthquakes.
Q.8	(B)	The Richter scale measures earthquake magnitude. Each whole-number increase represents roughly a 32-fold increase in released energy.
Q.9	(A)	A volcanic crater is the bowl-shaped depression or vent around the volcanic mouth through which eruptive materials are ejected.
Q.10	(A)	Weathering is the in-situ physical breakdown and chemical alteration of rocks at or near the Earth's surface without being transported away.
Q.11	(B)	Erosion involves the wearing away and dynamic transportation of rock particles by mobile natural agents like rivers, glaciers, wind, and sea waves.
Q.12	(A)	A gorge is a deep, narrow river valley with very steep rock walls formed by intensive vertical down-cutting in young mountainous stages.
Q.13	(B)	In plains, a river flows sluggishly and develops S-shaped looping bends called meanders due to lateral erosion on outer banks and deposition on inner banks.
Q.14	(A)	When a meander loop is cut off from the main river channel during floods, it leaves a curved, crescent-shaped lake known as an oxbow lake.
Q.15	(A)	A delta is a triangular alluvial deposit formed at the mouth of a river as velocity drops upon meeting a standing body of water (e.g., Sundarbans delta).
Q.16	(A)	Glaciers widen, deepen, and straighten pre-existing river valleys, carving steep-sided, flat-bottomed U-shaped valleys (glacial troughs).
Q.17	(A)	Moraines are accumulations of glacial till (unsorted rock debris) deposited at the margins, sides (lateral), or terminus (terminal) of a glacier.
Q.18	(A)	Wind carries sand particles predominantly near the ground (within 1 meter), eroding the lower trunk of rocks faster and leaving mushroom-shaped rocks.
Q.19	(A)	Barchans are crescent-shaped shifting sand dunes with horns pointing in the direction of the prevailing wind.
Q.20	(A)	Loess is an extensive unstratified deposit of windblown, silt-sized dust particles, creating rich, easily cultivated agricultural soils.

Q. No.	Correct	Curricular Reference & Conceptual Rationale
Q.21	(A)	Sea cliffs are steep vertical rock faces formed along coastlines where relentless wave action undermines and notches the base of coastal rocks.
Q.22	(A)	When sea caves in a headland unite from both sides or erode completely through, they form a natural bridge or sea arch.
Q.23	(A)	A sea stack is a steep, isolated vertical pillar of rock in the sea, left standing after the collapse of a sea arch roof.
Q.24	(A)	Beaches are coastal depositional landforms created when constructive waves wash sand and pebbles onto the shore.
Q.25	(A)	The Himalayas were formed by the continent-continent collision of the Indian Plate moving northwards into the Eurasian Plate at a convergent boundary.